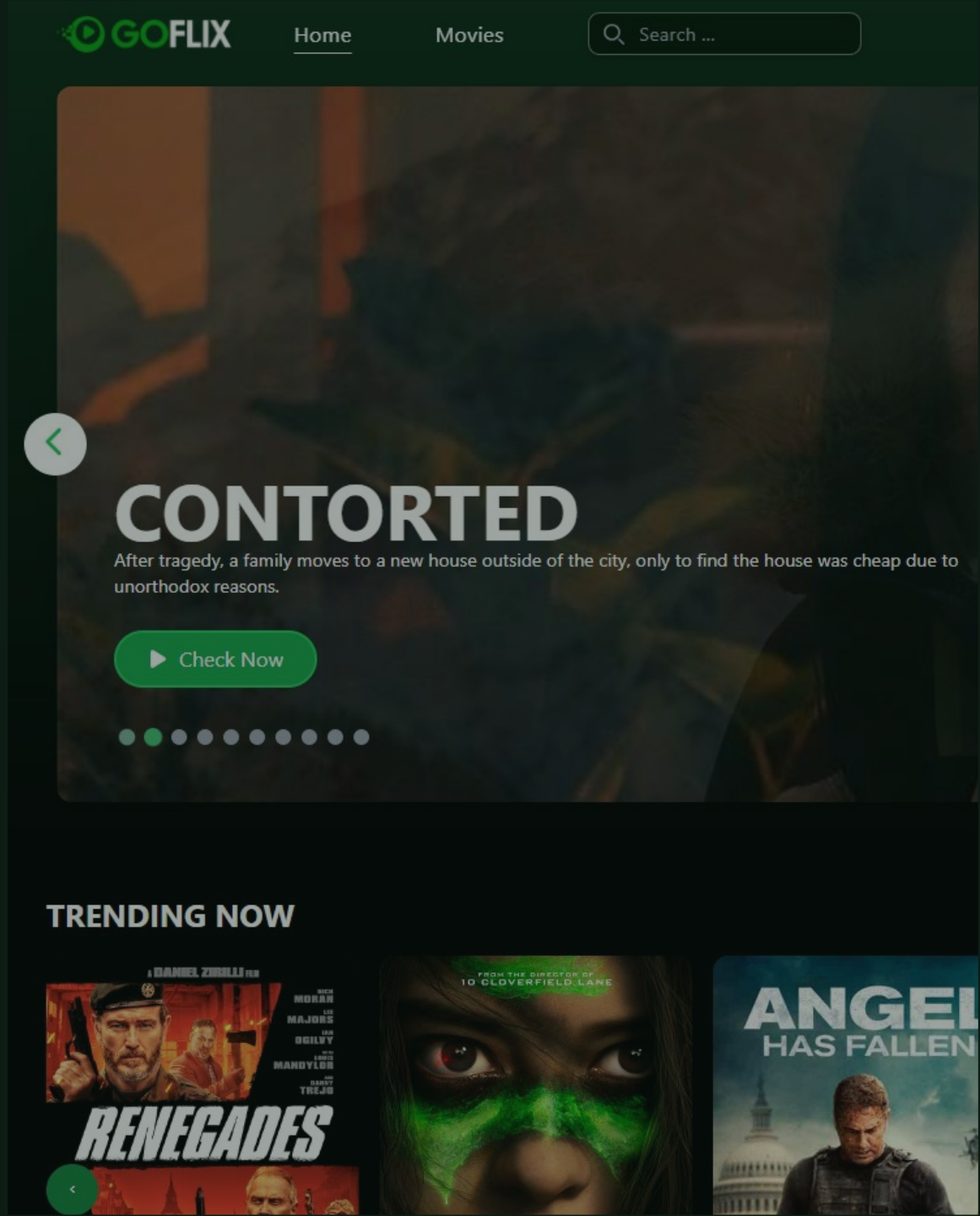
STREAMING PLATFORM

Goflix

The admin dashboard behind the catalog, the plans, and who may watch what

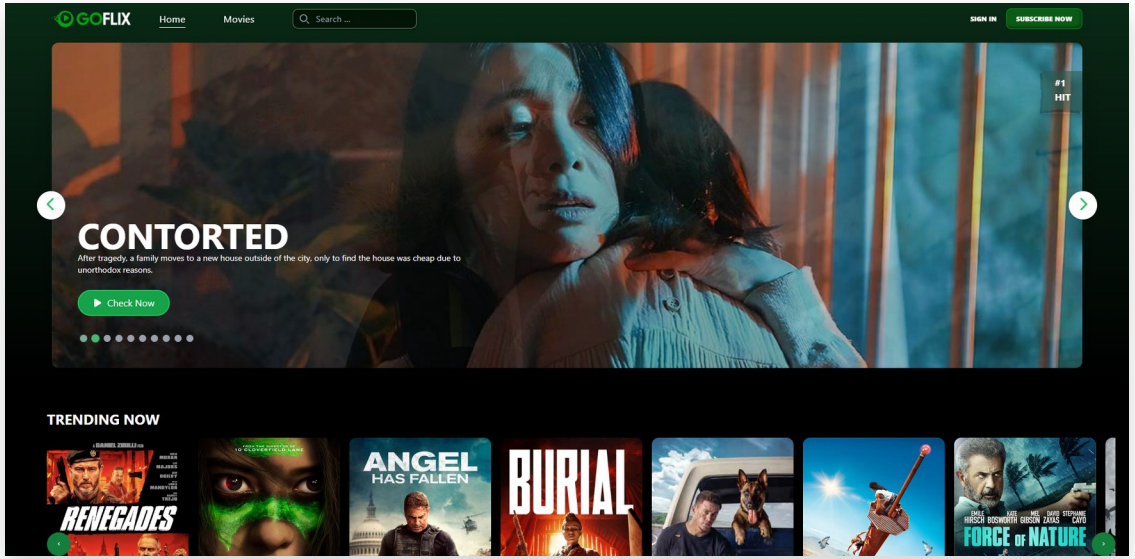
Enrico Saputra · Backend developer, Laravel dashboard

Client code and data are private. The demo rebuilds the Laravel dashboard on the real data model, with made-up data.

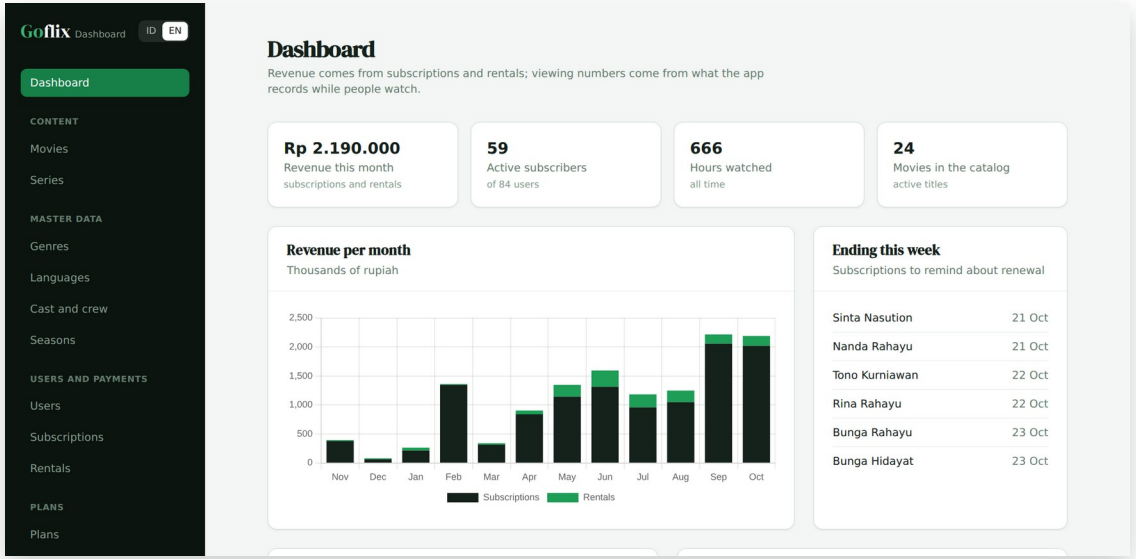


Viewers see this; the team works here

01



Viewer app · Vue + Go



Dashboard · Laravel · my part

The content team adds movies and episodes, the business team sets plans and coupons, and support answers "why can't I watch this?"

Three ways to watch

02



Free

Free titles play for anyone with an active account.



Plan

A monthly or yearly plan opens every premium title.



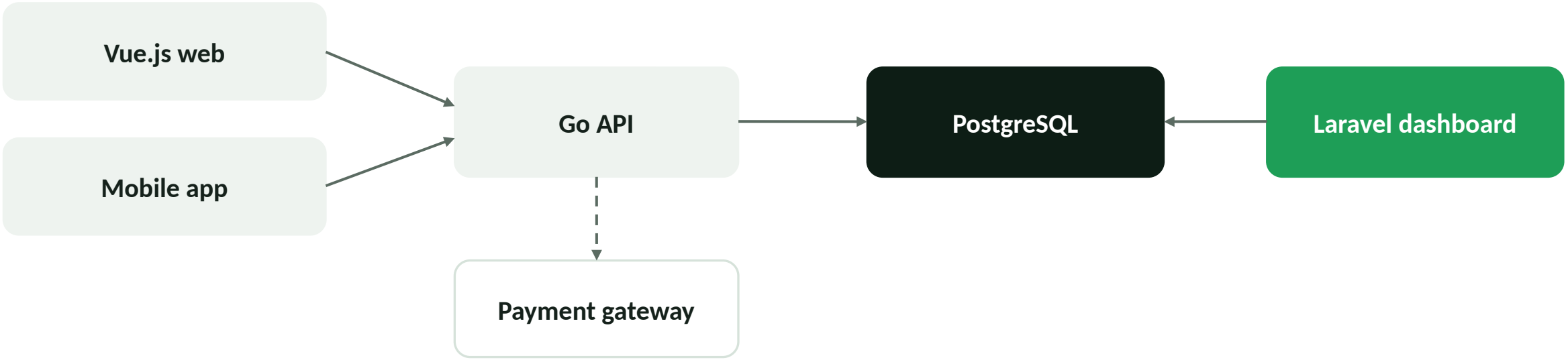
Rental

One premium movie, for a few days, without a plan.

My part: the catalog, plans and coupons, users and payments, and the reports on them.

Three parts, one database

03



Two writers

The API writes activity and in-app payments; the dashboard writes the catalog, plans and corrections.

The server decides

The app never decides access on its own. The dashboard shows support the same rule.

No API in the dashboard

Everything the apps need goes through the Go API.

The data model in three groups

04

Catalog

- movies
- shows
- episodes
- seasons
- genres
- languages
- people

Users and money

- users
- packages
- package_features
- subscriptions
- rentals
- coupons

Activity

- watch_history
- bookmarks

Written by the apps while people watch. The dashboard only reads it, for reports and continue watching.

users.subscription_ends_on is the source of truth for a fast access check; only SubscriptionService may change it.

Relations: from text to pivot tables

05

Production

```
video
  id: 12
  category_id: '6,12,19'
  cast_id: '1,108,109,110'
```

Cannot be joined or indexed; deleted ids stay behind.

Demo

```
genre_movie
  (movie_id 12, genre_id 6)
  (movie_id 12, genre_id 12)
  (movie_id 12, genre_id 19)
```

Foreign keys, normal joins, and the genre report is one query.

Gradual migration: add the pivot, fill it from the old column, move readers one by one, then drop the old column.

Renewing without losing paid days

06

Active until 20 Mar, buys 12 months on 10 Mar



Ended 31 Jan, buys 1 month on 10 Mar



One place

Only SubscriptionService moves users.subscription_ends_on.

One transaction

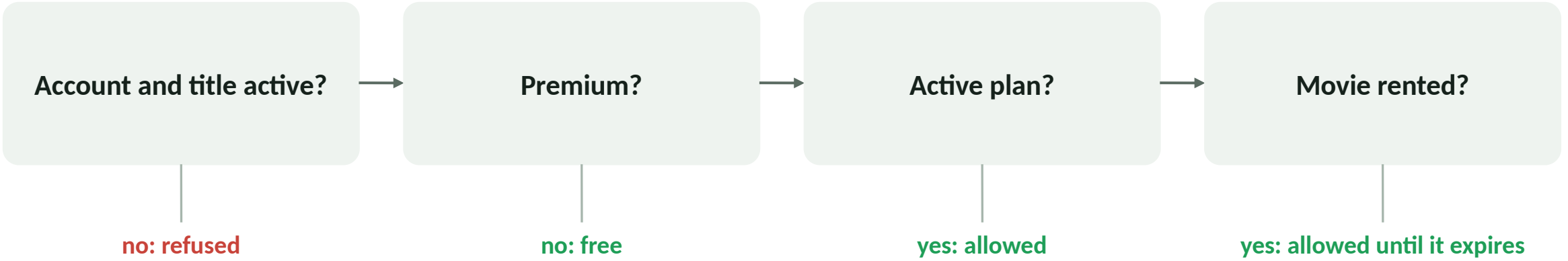
The payment and the end date are saved together, or neither is.

A row lock

lockForUpdate makes a second payment wait and count from the new date.

Who may watch what

07



Refused: needs a plan, or a rental if the movie has a rent price

An episode is premium on its own or through its series. One AccessService is used everywhere, and the dashboard shows it per user so support can answer complaints.

Rentals and coupons

08

Renting one movie

- Only premium movies with a rent price
- Lasts rent_days days from payment
- Active subscribers cannot rent: it is already included
- The same movie cannot be rented twice at once

Coupons

- Fixed amount or percentage
- Only valid between its start and end dates
- Never takes a price below zero
- A single-use coupon works once per user

Example: an active subscriber tries to rent a premium movie → refused, because the plan already includes it. Nobody pays twice, and refund requests drop.

Decisions worth talking about

09

1 End date on users

Deliberate duplication: one indexed column beats finding the latest subscription on every play.

2 Payments cannot be edited

Subscriptions and rentals are financial records. A correction is a new entry.

3 Deactivate, not delete

A genre, plan, coupon or user in use can only be deactivated.

4 Episodes updated in place

The same rows are kept, so viewing history stays attached.

Tech stack and why

10

Laravel 10

Admin dashboard: forms, validation, CRUD. The part I built.

PostgreSQL

One database for the API and the dashboard, with transactions and row locks.

Go

The apps' API: heavy playback traffic, low latency.

Vue.js

The viewer web app.

Demo: Laravel 10, Blade, Tailwind, Chart.js and PostgreSQL, like production. The green one is my part.

Keeping the code easy to read

11

```
return DB::transaction(function () use (...) {
    $user = User::lockForUpdate()
        ->findOrFail($user->id);
    $startsOn = $user->subscription_ends_on
        ?->gte($today)
        ? $user->subscription_ends_on->addDay()
        : $today;
    $endsOn = $startsOn->copy()
        ->addMonthsNoOverflow($months)->subDay();
    $user->subscriptions()->create([...]);
    $user->update(['subscription_ends_on' => $endsOn]);
});
```

Rules in services

Thin controllers; plans, rentals, coupons and access are plain classes.

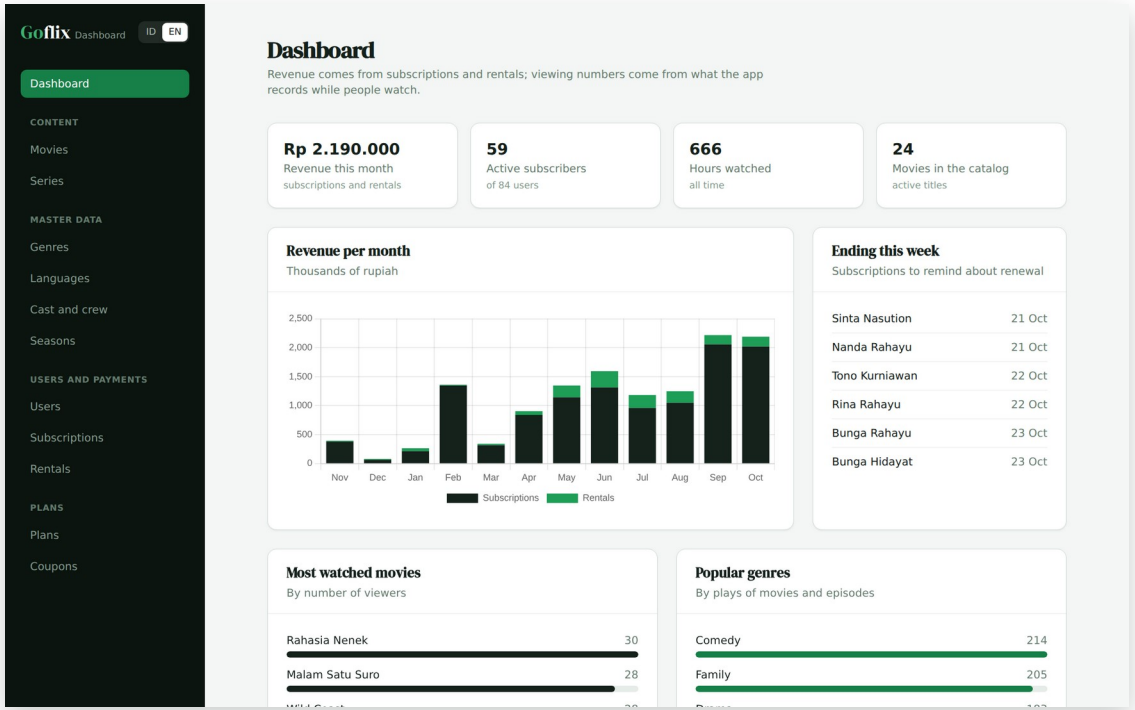
Enums, not numbers

Sign-in, role and coupon type are typed.

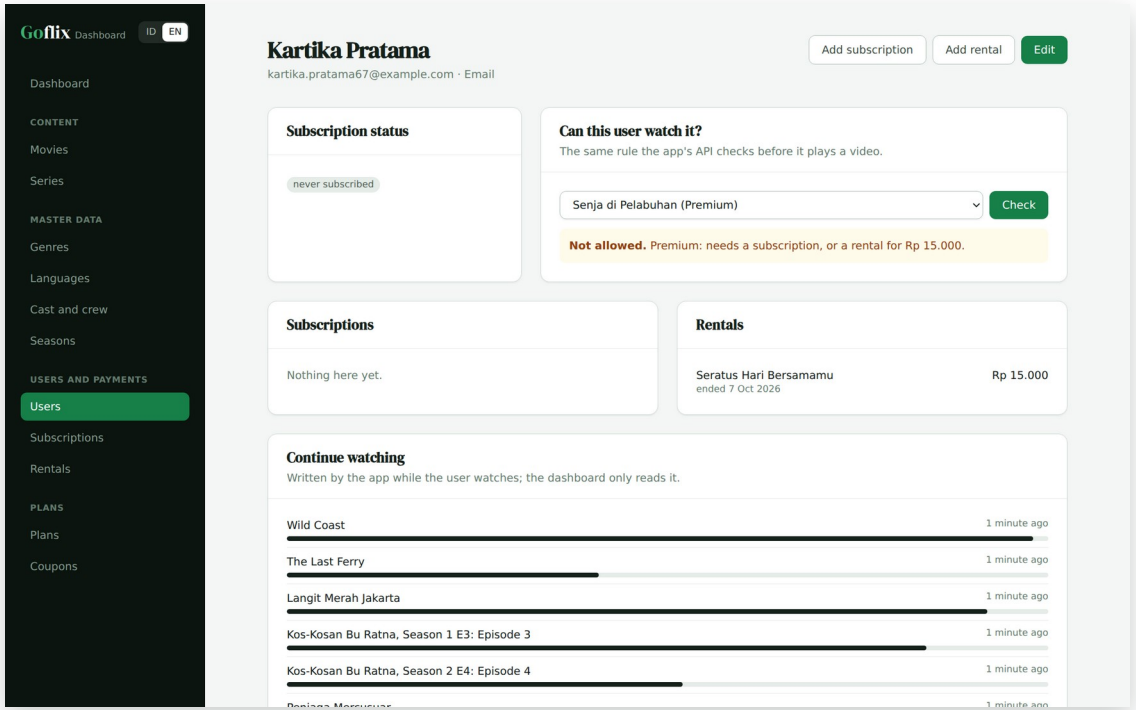
Tests on the rules

33 feature tests.

The demo app



Dashboard: plan and rental revenue per month



User page: status, payments and the access check

What I would improve

13

Move to pivot tables	Replace the comma-separated ids step by step, with no downtime.
One shared rule contract	The access rule lives in Go and Laravel; one shared set of test cases keeps them in step.
Large viewing history	Partition watch_history by month, report from a daily summary table.
Constraints and indexes	Foreign keys and indexes on the access-check and report columns.

Try it yourself

Live demo	https://goflix-dashboard-demo.vercel.app
Slides (PDF)	https://goflix-dashboard-demo.vercel.app/goflix-walkthrough.pdf
Slides (PPTX)	https://goflix-dashboard-demo.vercel.app/goflix-walkthrough.pptx

The demo code is in a private GitHub repo; access is given during the interview. The dashboard speaks Bahasa Indonesia and English; demo data is restored every 3 hours.